

Introduction to Computer Vision (CS642)

Lecture 05 - Edge Detection

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- 1 Edge detection
 - Overview
 - Discontinuity in Images
- 2 Edges Characterization
- 3 Effect of noise over edges
- 4 Edge detectors
 - Prewit and Sobel Edge Detector
 - Marr Hildreth Edge Detector
- 5 Canny Edge Detector
 - Non-Maximum Suppression
 - Hysteresis Threshold

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Edge detection

Overview



Figure: Edge detection examples. Reprinted from Alper Yilmaz, Mubarak Shah Fall 2012UCF.

Edge detection

Overview

- Edges are the sudden changes or **discontinuities** in an image
 - Intuitively, most semantic and **shape information** from the image can be encoded in the edges
 - More compact than pixels



Figure: Edges in image. Reprinted from D. Lowe.

Edge detection

Overview

- Edges are used to **extract information** to recognize the objects.
- Recover **geometry** and **view points**.

Edge detection

Overview

- Edges are caused by variety of reasons.

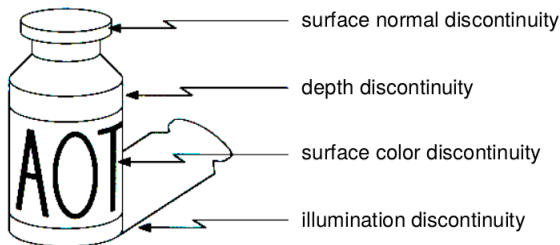


Figure: Edges are caused by variety of factors.

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Edge detection

Discontinuities in images

- The **discontinuities** in images cause edges.

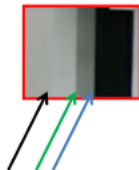


Figure: Discontinuities in image color intensities. Reprinted from D. Hoiem.

Edge detection

Discontinuities in images

- The **discontinuities** in images cause edges.



Source

Figure: Discontinuities in image color intensities. Reprinted from D. Hoiem.

Edge detection

Discontinuities in images

- Edges are caused by variety of reasons.



Source

Figure: Discontinuities in image color intensities. Reprinted from D. Hoiem.

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Characterizing Edges

- Edges are the rapid change in the image intensity values.

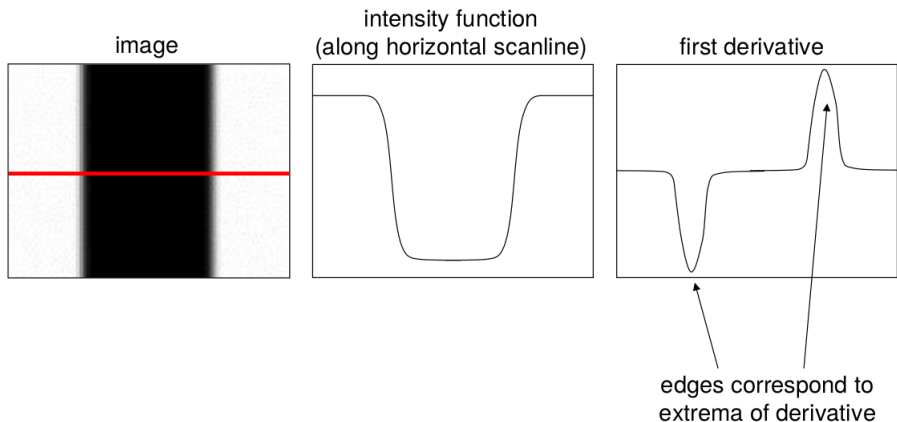


Figure: Change in color intensity. Derivative shows the change.

Characterizing Edges

Color Intensity and derivative (change)

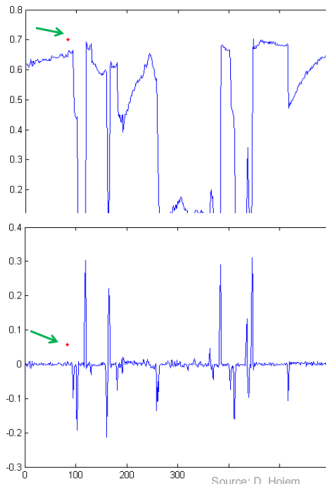
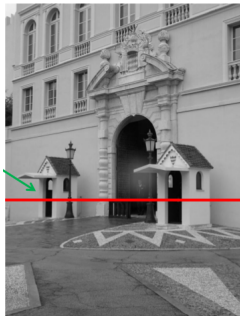


Figure: Image color intensity. Source D. Hoiem.

Characterizing Edges

With some Gaussian noise

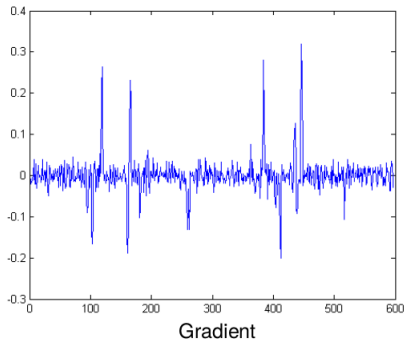
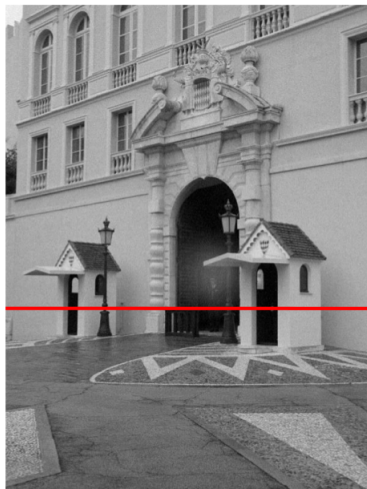


Figure: Color intensity with some Gaussian noise. Source D. Hoiem.

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Effect of noise over edges

- Plotting the image intensity.

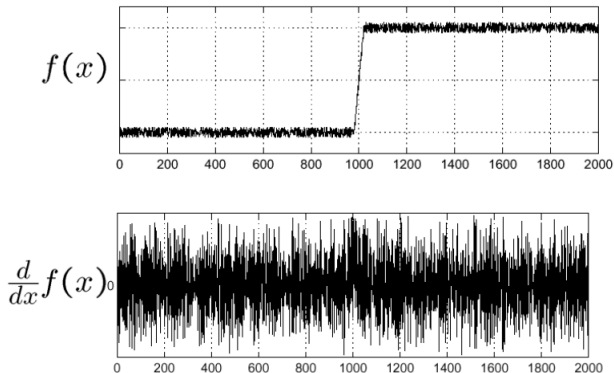


Figure: Color intensity with some Gaussian noise. Source D. Hoiem.

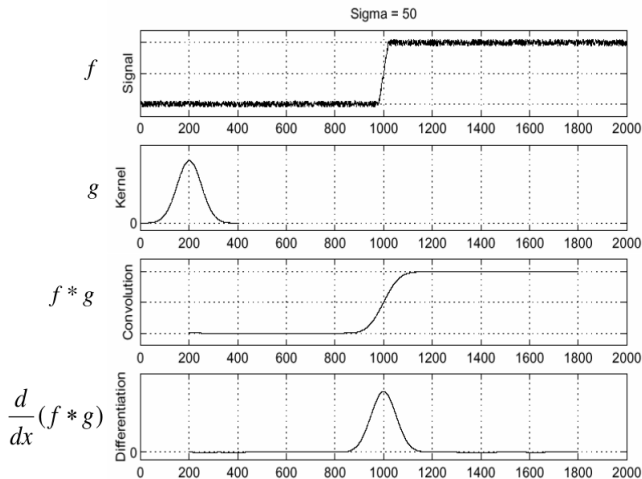
Effect of noise over edges

- Difference filters respond strongly to noise
 - Image noise results in pixels that look very different from their neighbors
 - Generally, the larger the noise the stronger the response
- What can we do about it?

Effect of noise over edges

Smooth the image first

- Smooth the image first and look for the peaks in $\frac{d}{dx}(f * g)$



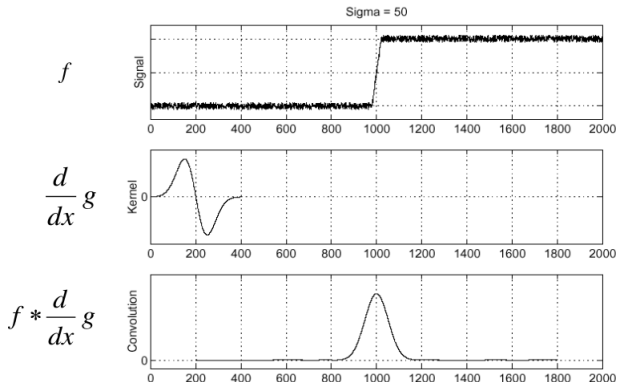
Effect of noise over edges

Derivation for the Convolution

- Differentiation is convolution, and convolution is **associative**:

$$\frac{d}{dx}(f * g) = f * \frac{d}{dx}g$$

- It reduces one operation.



An Edge Detector

- Criteria of a good edge detector.
 - **Good detection:** the optimal detector should find all real edges, ignoring noise or other artifacts
 - **Good localization:**
 - The edges detected must be as close as possible to the true edges
 - The detector must return one point only for each true edge point

Edge detection

Detecting Discontinuities

- Image derivatives

$$\frac{\partial f}{\partial x} = \lim_{\varepsilon \rightarrow 0} \left(\frac{f(x + \varepsilon) - f(x)}{\varepsilon} \right) \rightarrow \frac{\partial f}{\partial x} \approx \frac{f(x_{n+1}) - f(x)}{\Delta x}$$

- Convolve image with derivative filters

Backward difference $[-1 \ 1]$

Forward difference $[1 \ -1]$

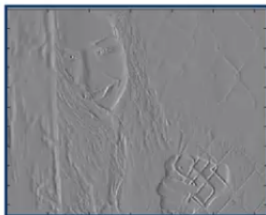
Central difference $[-1 \ 0 \ 1]$

Edge detection

Image Derivatives



Image I



$$I_x = I * \begin{bmatrix} 1 & -1 \end{bmatrix}$$

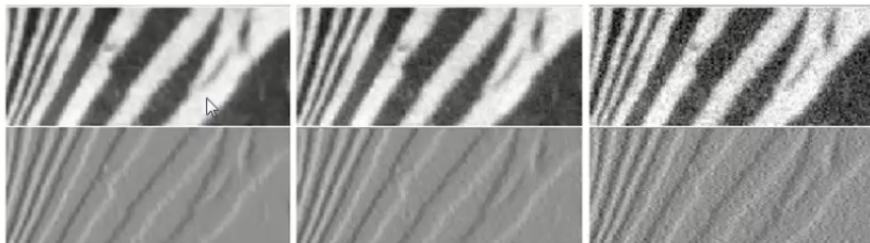


$$I_y = I * \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Figure: Source: Yilmaz, Mubarak Shah

Edge detection

Derivatives and Noise



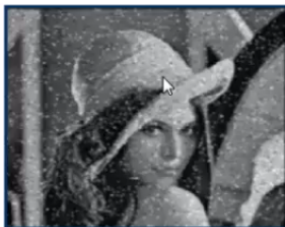
Increasing noise $\longrightarrow$

Figure: Zero mean additive Gaussian noise. Source: Yilmaz, Mubarak Shah

Edge detection

Gaussian Smoothing

- Scale of Gaussian σ .
 - As σ increases, more pixels are involved in average.
 - As σ increases, image is more blurred.
 - As σ increases, noise is more effectively suppressed.

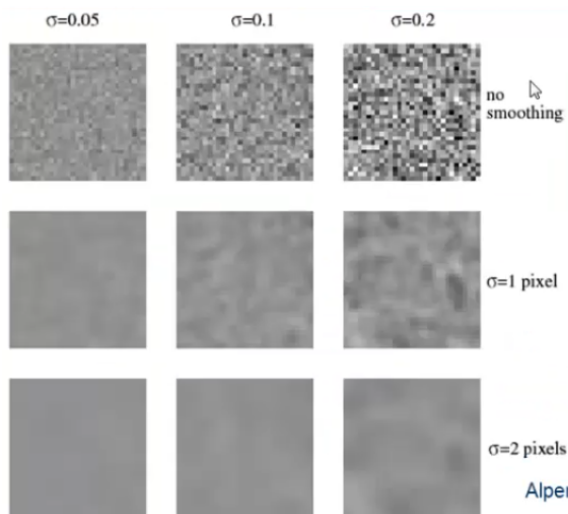


$$g(x, y) = e^{\frac{-(x^2 + y^2)}{2\sigma^2}}$$

Figure: Source: Yilmaz, Mubarak Shah

Edge detection

Gaussian Smoothing (Examples)



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- Gradient operators
 - Prewit
 - Sobel
- Laplacian of Gaussian (Marr-Hildreth)
- Gradient of Gaussian (Canny)

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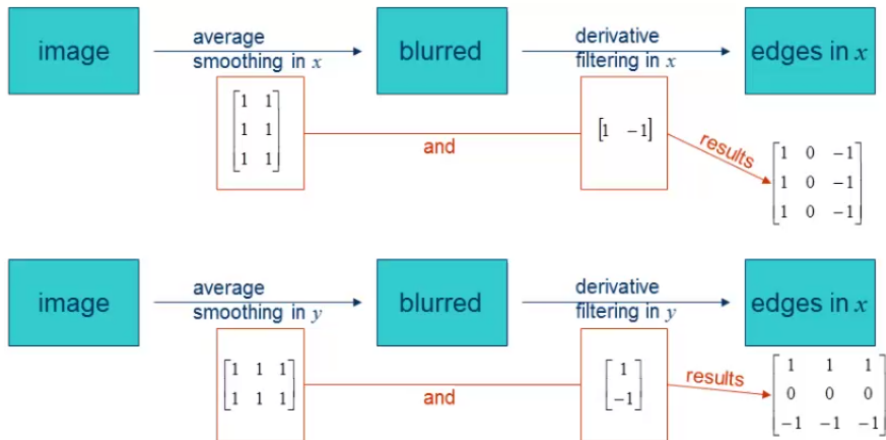
Edge detectors

Prewit and Sobel Edge Detector

- Compute derivatives
 - in x and y directions
- Find gradient magnitude
- Threshold gradient magnitude

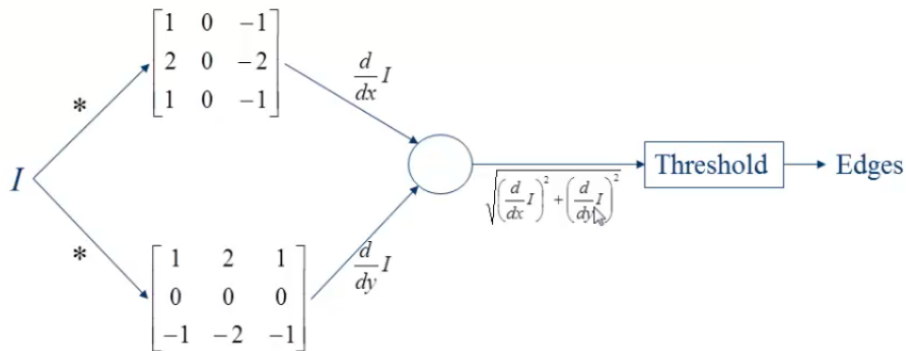
Edge detectors

Prewitt Edge Detector



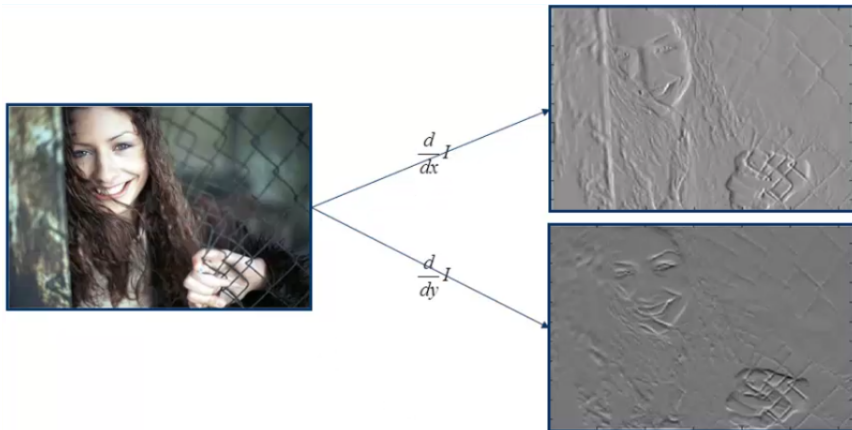
Edge detectors

Sobel Edge Detector



Edge detectors

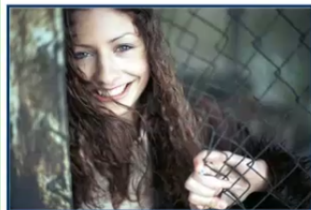
Sobel Edge Detector



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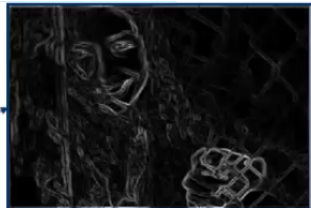
Edge detectors

Sobel Edge Detector



$$\Delta = \sqrt{\left(\frac{d}{dx}I\right)^2 + \left(\frac{d}{dy}I\right)^2}$$

$$\Delta \geq Threshold = 100$$



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Edge detectors

Marr Hildreth edge detector

- Smooth image by Gaussian filter to produce image S
- Apply Laplacian to S
 - Used in mechanics, electromagnetics, wave theory, quantum mechanics and Laplace equation
- Find **zero crossings**
 - Scan along each row, record an edge point at the location of zero-crossing.
 - Repeat above step along each column

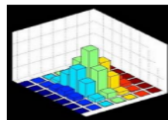
Edge detectors

Marr Hildreth edge detector

• Gaussian smoothing

$$\underbrace{\text{smoothed image}}_{\hat{S}} = \underbrace{\text{Gaussian filter}}_{g} * \underbrace{\text{image}}_{I}$$

$$g = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{x^2+y^2}{2\sigma^2}}$$



• Find Laplacian

$$\Delta^2 S = \underbrace{\frac{\partial^2}{\partial x^2}}_{\substack{\text{second derivative} \\ \text{order in } x}} S + \underbrace{\frac{\partial^2}{\partial y^2}}_{\substack{\text{second derivative} \\ \text{order in } y}} S$$

- ∇ is used for gradient (derivative)
- Δ is used for Laplacian

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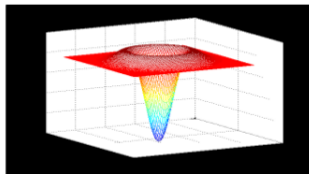
Edge detectors

Marr Hildreth edge detector

- Deriving the Laplacian of Gaussian (LoG)

$$\Delta^2 S = \Delta^2 (g * I) = (\Delta^2 g) * I$$

$$\Delta^2 g = -\frac{1}{\sqrt{2\pi}\sigma^3} \left(2 - \frac{x^2 + y^2}{\sigma^2} \right) e^{-\frac{x^2 + y^2}{2\sigma^2}}$$



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Edge detectors

Finding Zero Crossing

- Four cases of zero-crossings
 - $\{+,-\}$
 - $\{+,0,-\}$
 - $\{-,+\}$
 - $\{-,0,+\}$
- Slope of zero-crossing $\{a, -b\}$ is $|a+b|$.
- To mark an edge
 - Compute slope of zero-crossing
 - Apply a threshold to slope

Finding Zero Crossing

Example

I



$I * (\Delta^2 g)$



Zero crossings of $\Delta^2 S$



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Marr Hildreth edge detector

Algorithm

- Compute LoG

- Use 2D filter

$$\Delta^2 g(x, y)$$

- Use 4 1D filter

$$g(x), g_{xx}(x), g(y), g_{yy}(y)$$

- Find zero-crossings from each row
- Find slope of zero-crossings
- Apply threshold to slope and mark edges

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Canny Edge detector

Quality of an edge

- **Good Detection:** The optimal detector must minimize the probability of false positives as well as false negatives.
- **Good Localization:** The edges detected must be as close as possible to the true edges.
- **Single Response Constraint:** The detector must return one point only for each edge point.

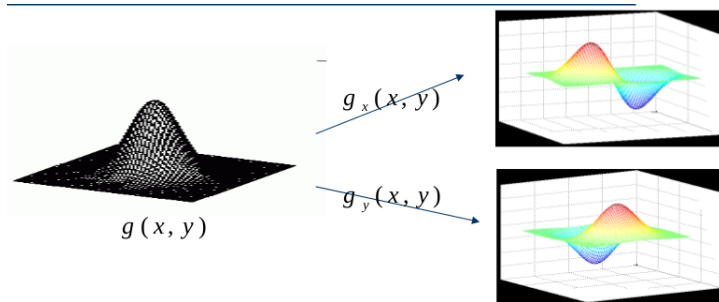
Canny Edge detector

Algorithm steps

- 1 Smooth image with Gaussian filter
- 2 Compute derivative of filtered image
- 3 Find magnitude and orientation of gradient
- 4 Apply “Non-maximum Suppression”
- 5 Apply “Hysteresis Threshold”

Canny Edge Detector

Derivative of Gaussian



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Canny Edge detector

Smoothing and Derivatives

1 Smoothing

$$S = I * g(x, y) = g(x, y) * I$$

2 Derivate

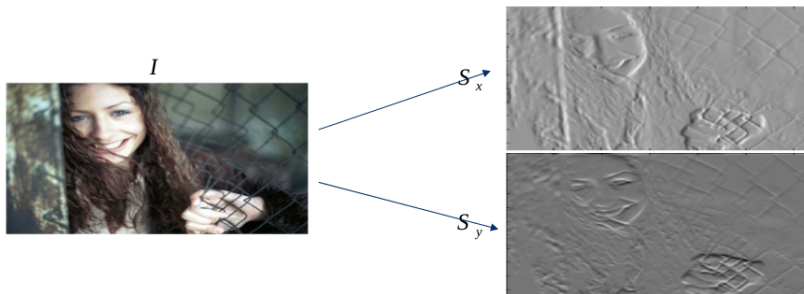
$$\nabla S = \nabla (g * I) = (\nabla g) * I$$

$$\nabla S = \begin{bmatrix} g_x \\ g_y \end{bmatrix} * I = \begin{bmatrix} g_x * I \\ g_y * I \end{bmatrix}$$

$$\nabla g = \begin{bmatrix} \frac{\partial g}{\partial x} \\ \frac{\partial g}{\partial y} \end{bmatrix} = \begin{bmatrix} g_x \\ g_y \end{bmatrix}$$

Canny Edge detector

Smoothing and Derivatives



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Canny Edge detector

Gradient Magnitude and Direction

- Computer gradient magnitude and gradient direction

(S_x, S_y) Gradient Vector

$$\text{magnitude} = \sqrt{(S_x^2 + S_y^2)}$$

$$\text{direction} = \theta = \tan^{-1} \frac{S_y}{S_x}$$



image



gradient magnitude

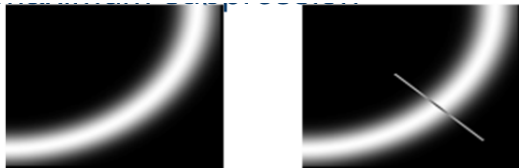
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Canny Edge detector

Non Maximum Suppression



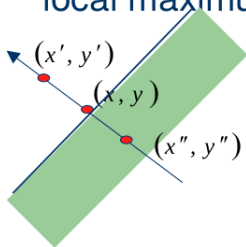
We wish to mark points along the curve where the **magnitude is largest**. We can do this by looking for a maximum along a slice normal to the curve (non-maximum suppression). These points should form a curve. There are then two algorithmic issues: at which point is the maximum, and where is the next one?

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Canny Edge detector

Non Maximum Suppression

- Suppress the pixels in $|\nabla S|$ which are not local maximum



$$M(x, y) = \begin{cases} |\nabla S|(x, y) & \text{if } |\nabla S|(x, y) > |\Delta S|(x', y') \\ & \& |\Delta S|(x, y) > |\Delta S|(x'', y'') \\ 0 & \text{otherwise} \end{cases}$$

x' and x'' are the neighbors of x along normal direction to an edge

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Canny Edge detector

Non Maximum Suppression

$$|\Delta S| = \sqrt{S_x^2 + S_y^2}$$



M



For visualization

$M \geq \text{Threshold} = 25$



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Canny Edge detector

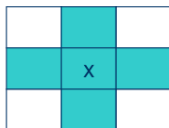
Hysteresis Threshold

- If the gradient at a pixel is
 - above **High**, declare it as an 'edge pixel'
 - below **Low**, declare it as a "non-edge-pixel"
 - **between** "low" and "high", Consider its neighbors iteratively then declare it an "edge pixel" if it is **connected** to an 'edge pixel' directly or via pixels between "low" and "high".

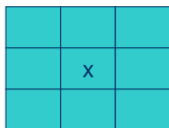
Canny Edge detector

Hysteresis Threshold

- Connectedness



4 connected



8 connected

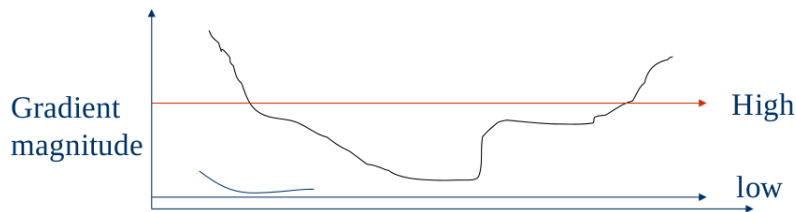


6 connected

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Canny Edge detector

Hysteresis Threshold



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Canny Edge detector

Hysteresis Threshold

- Scan the image from left to right, top-bottom.
 - The gradient magnitude at a pixel is above a high threshold declare that as an edge point
 - Then recursively consider the neighbors of this pixel.
“If the gradient magnitude is above the low threshold declare that as an edge pixel.”

Canny Edge detector

Hysteresis Threshold

M



regular

$M \geq 25$



Hysteresis

High = 35

Low = 15



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The End